IsdaSafe

Pond Water Quality Monitoring System

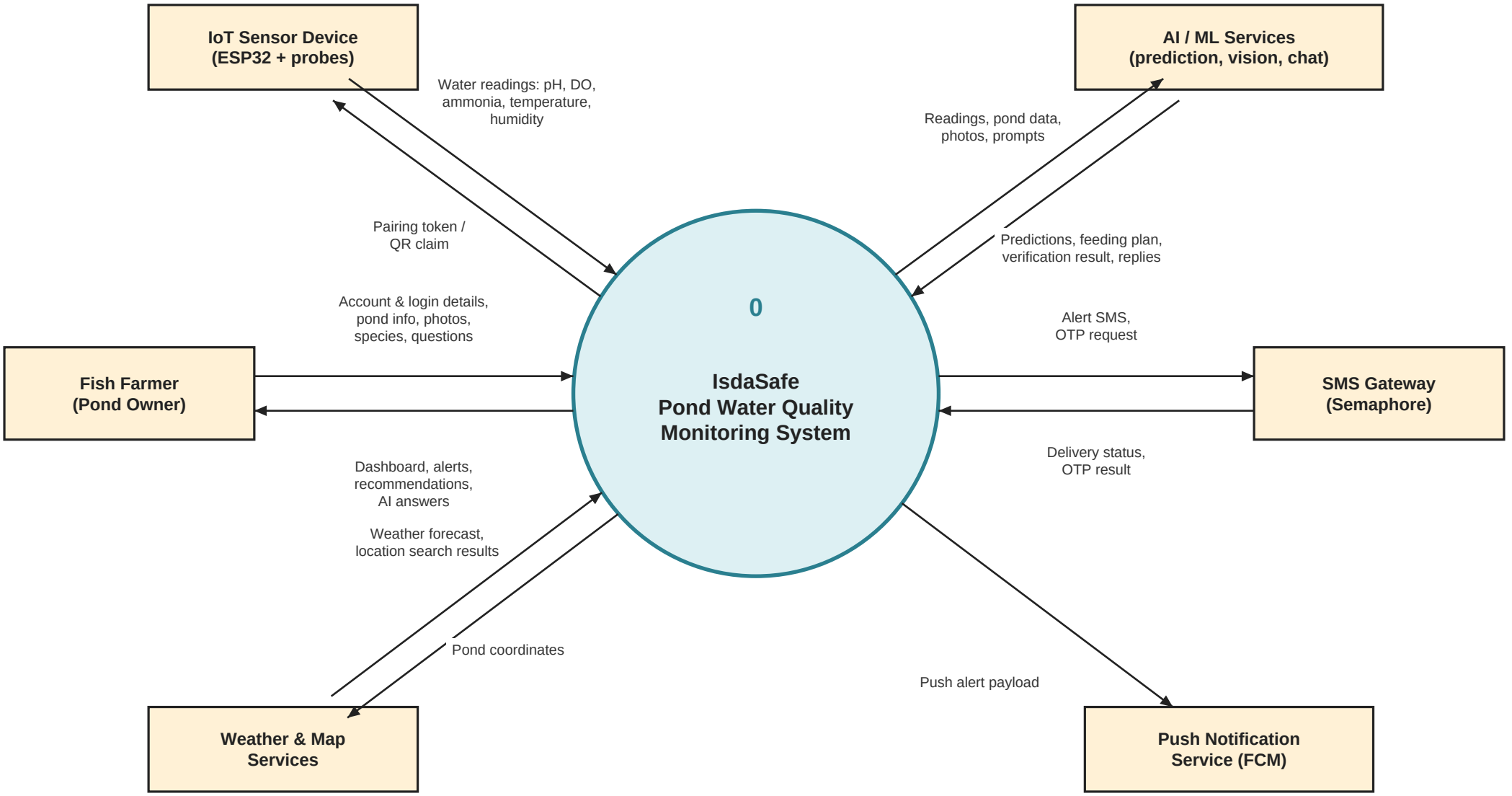
Example system analysis & design diagrams

1. **Context Diagram**
   The whole system as one process and the outside parties it talks to.
2. **Level 0 Data Flow Diagram**
   The system broken into its five major processes and data stores.
3. **Child Diagrams (Processes 1.0 - 5.0)**
   Each Level 0 process exploded into its sub-processes.
4. **State Transition Diagrams**
   Life cycle of pond verification and of a water-quality alert.
5. **Flowchart**
   The whole system step by step: sign-up, pond verification, pairing, monitoring, alerts, AI advice.
6. **Entity Relationship Diagram**
   The database tables (Supabase / PostgreSQL) and how they relate.
7. **Data Dictionary**
   Every column of every table: type, nullability, keys, default and meaning.
8. **Software Block Diagram**
   The software split into its major blocks: app, cloud backend, device firmware and outside services.
9. **Software Schematic Diagram**
   Every app module, backend function and service wired together on three sheets joined by off-page connectors, plus an interface list.

Context diagram: Yourdon/DeMarco notation. Level 0 and child DFDs: Gane & Sarson notation. ERD: crow's-foot notation. Block & schematic diagrams: software architecture view.

1. Context Diagram

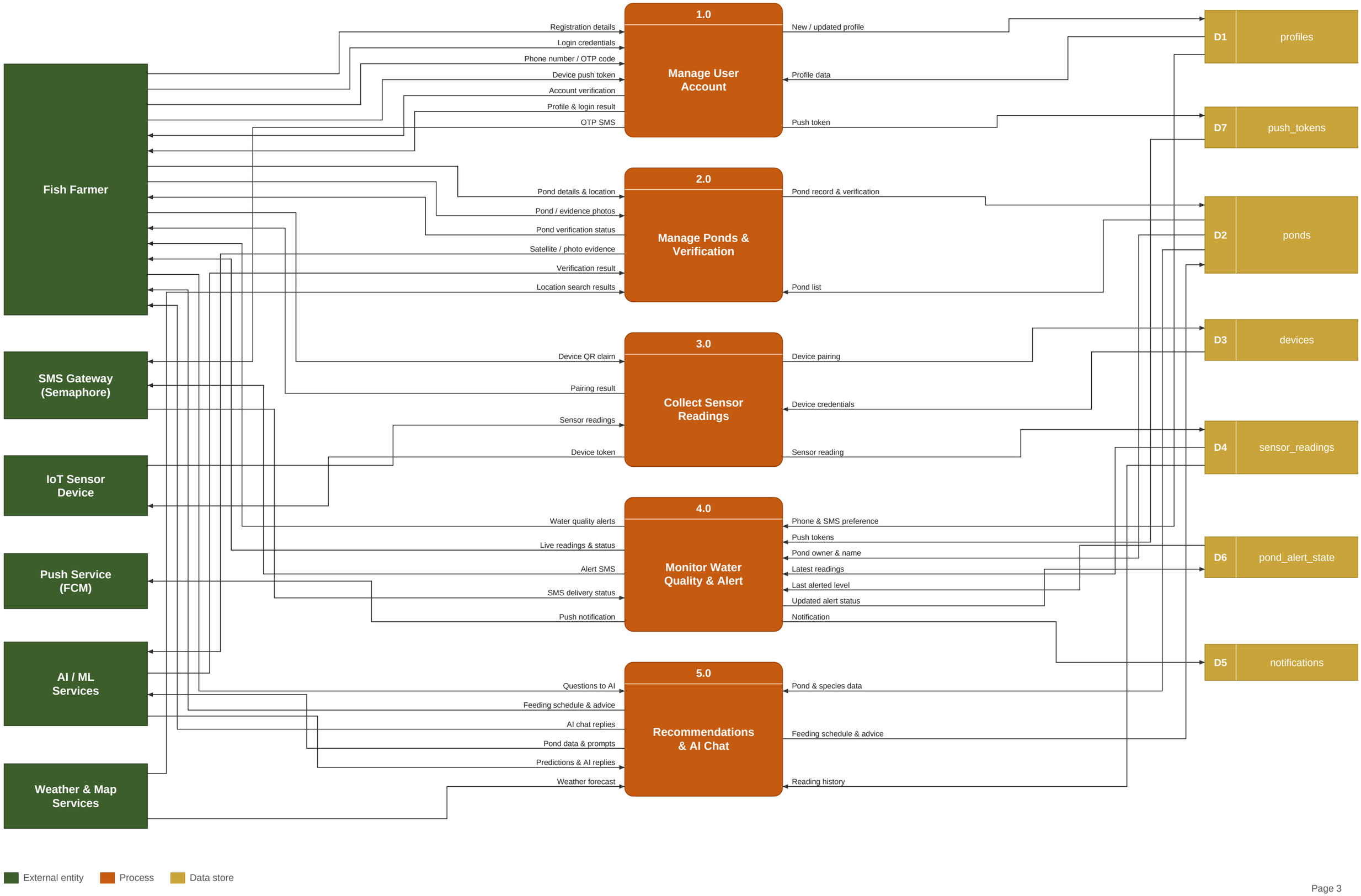
Shows IsdaSafe as a single process (0) and every external entity that sends data to it or receives data from it.



External entity | Process | Data store

2. Level 0 Data Flow Diagram

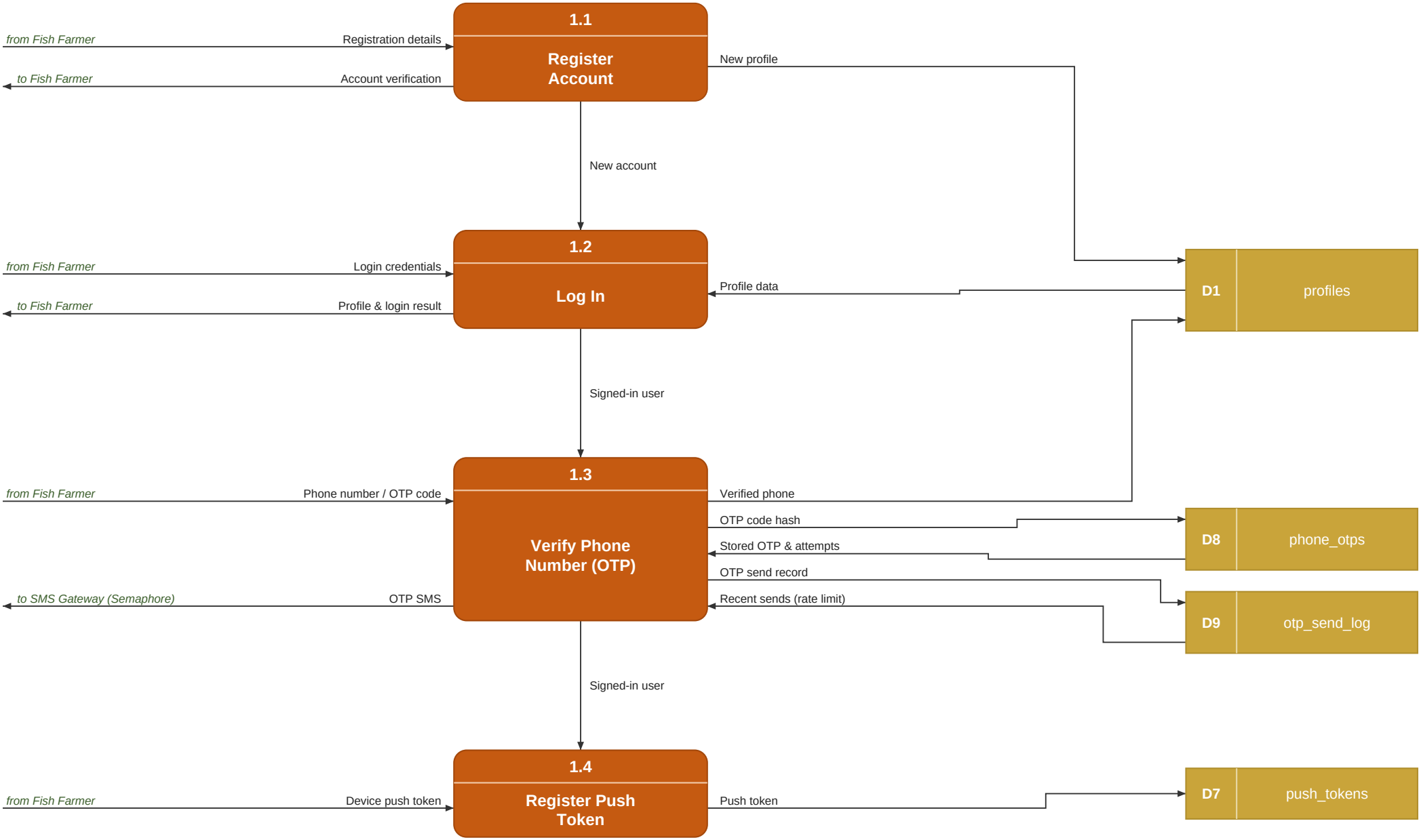
Process 0 exploded into its five major processes (Gane & Sarson notation). Green = external entity, orange = process, gold = data store. Every line is one data flow.



External entity Process Data store

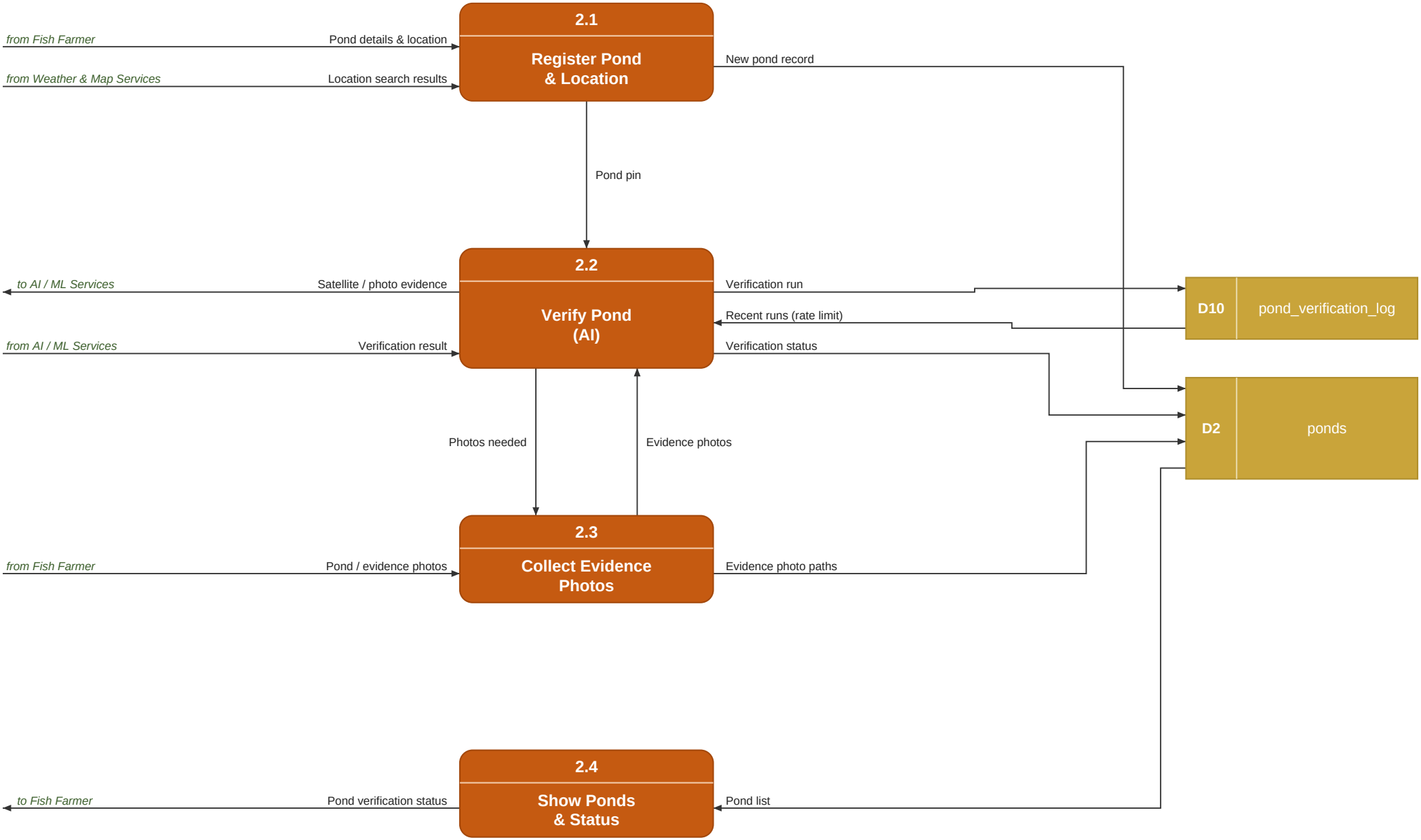
3. Child Diagram - Process 1.0: Manage User Account

Explodes process 1.0 of the Level 0 DFD. External entities are not drawn: each arrow from the left edge names its entity at the open end. Local stores D8, D9 appear first at this level.



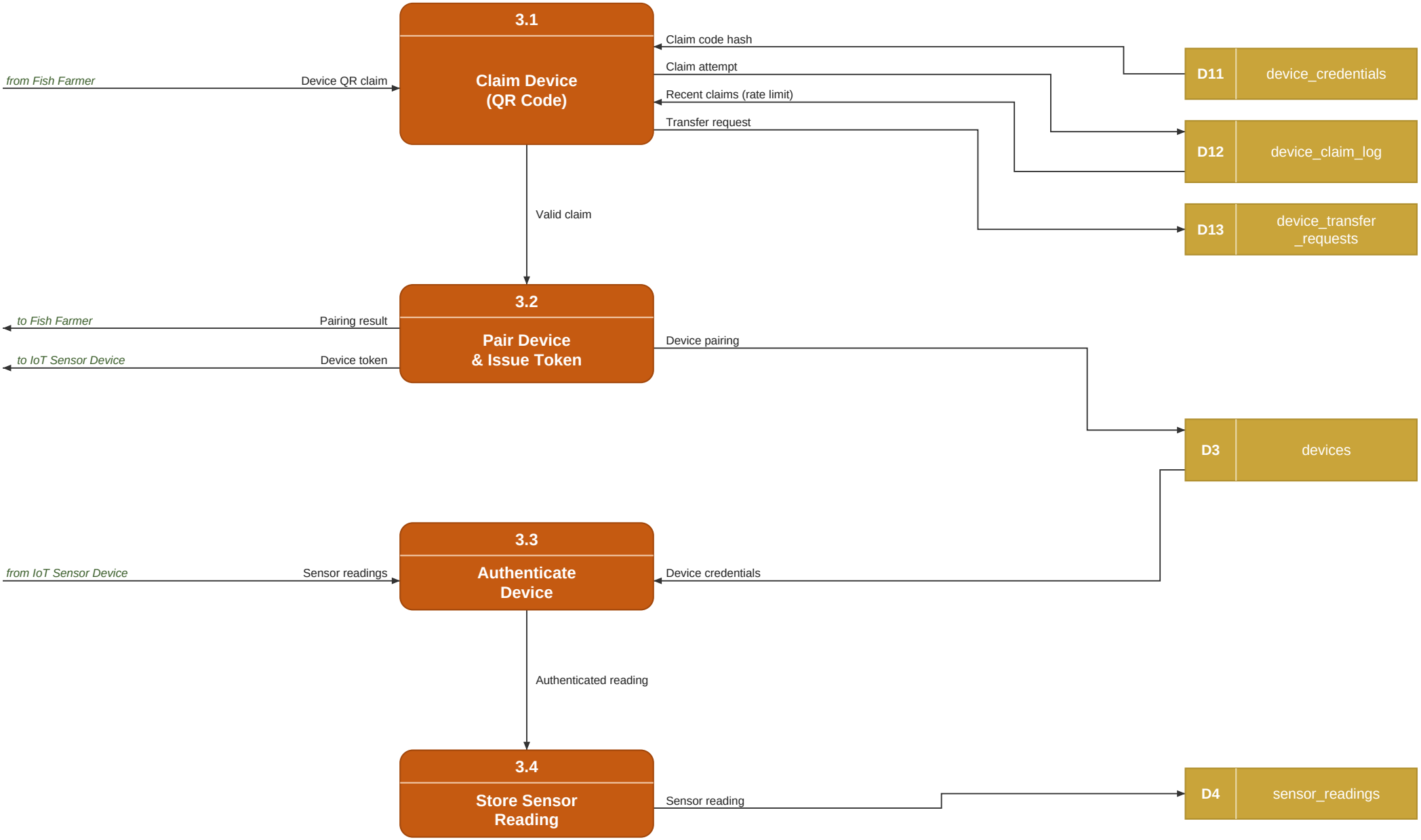
3. Child Diagram - Process 2.0: Manage Ponds & Verification

Explodes process 2.0 of the Level 0 DFD. External entities are not drawn: each arrow from the left edge names its entity at the open end. Local store D10 appears first at this level.



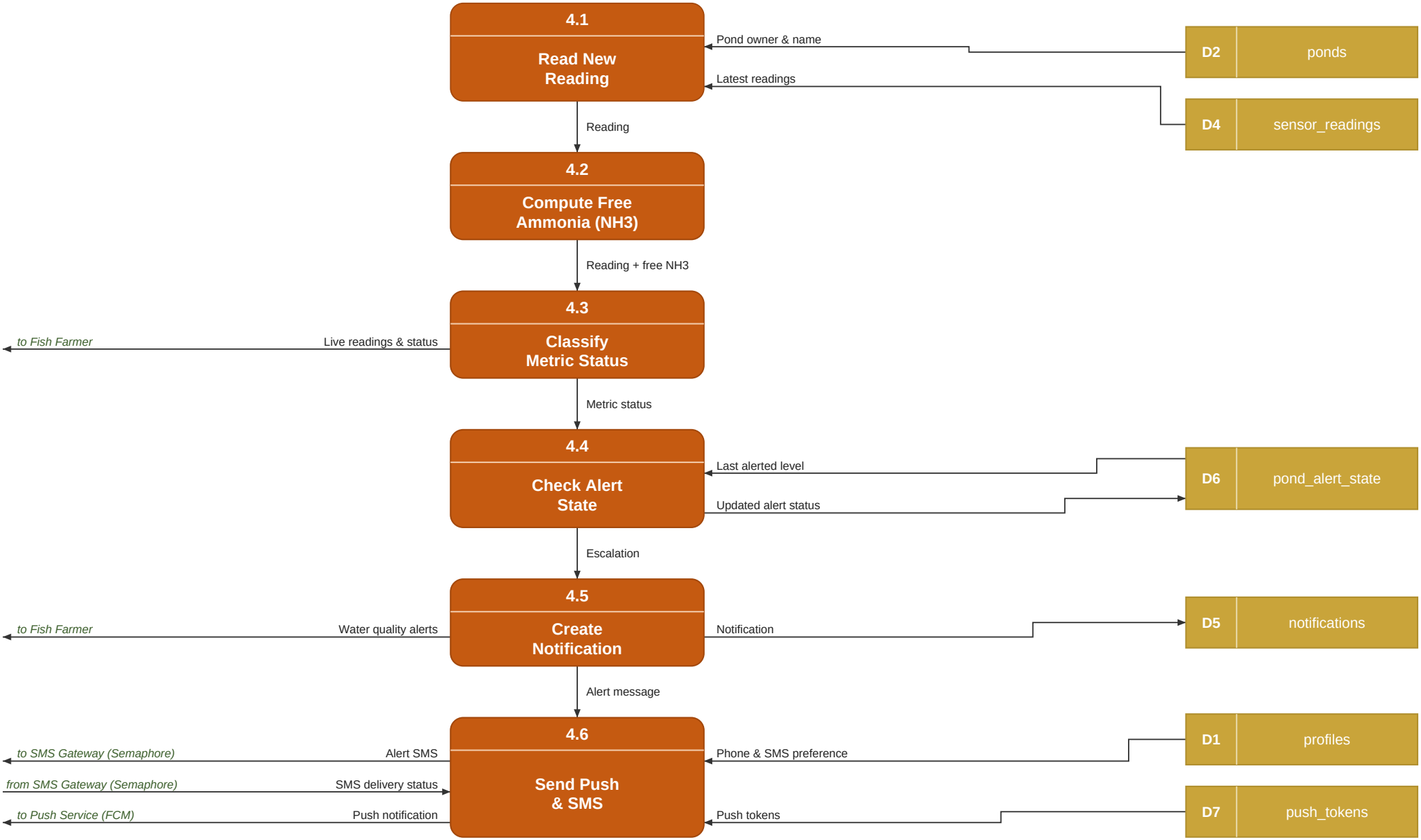
3. Child Diagram - Process 3.0: Collect Sensor Readings

Explodes process 3.0 of the Level 0 DFD. External entities are not drawn: each arrow from the left edge names its entity at the open end. Local stores D11, D12, D13 appear first at this level.



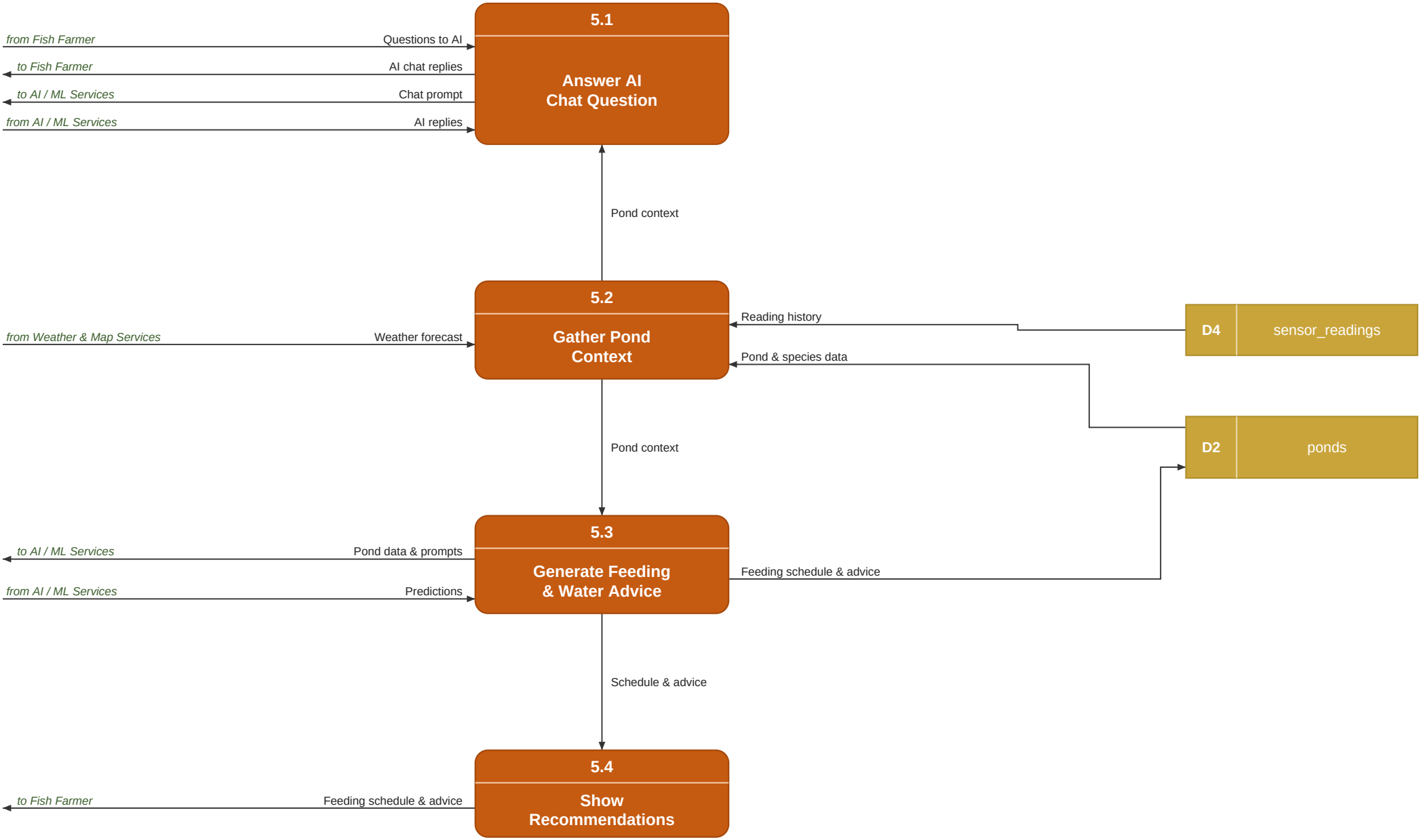
3. Child Diagram - Process 4.0: Monitor Water Quality & Alert

Explodes process 4.0 of the Level 0 DFD. External entities are not drawn: each arrow from the left edge names its entity at the open end.



3. Child Diagram - Process 5.0: Recommendations & AI Chat

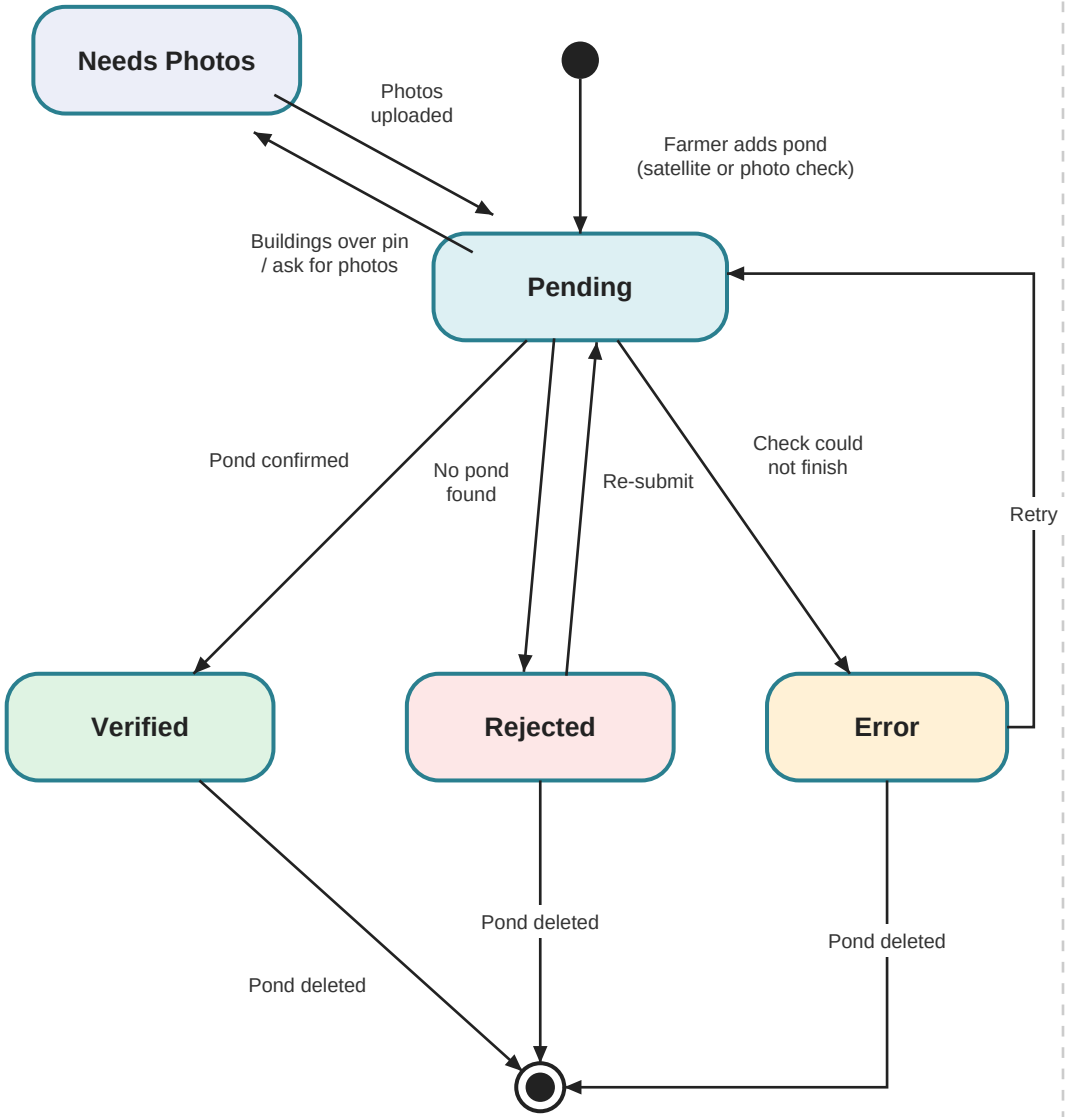
Explodes process 5.0 of the Level 0 DFD. External entities are not drawn: each arrow from the left edge names its entity at the open end.



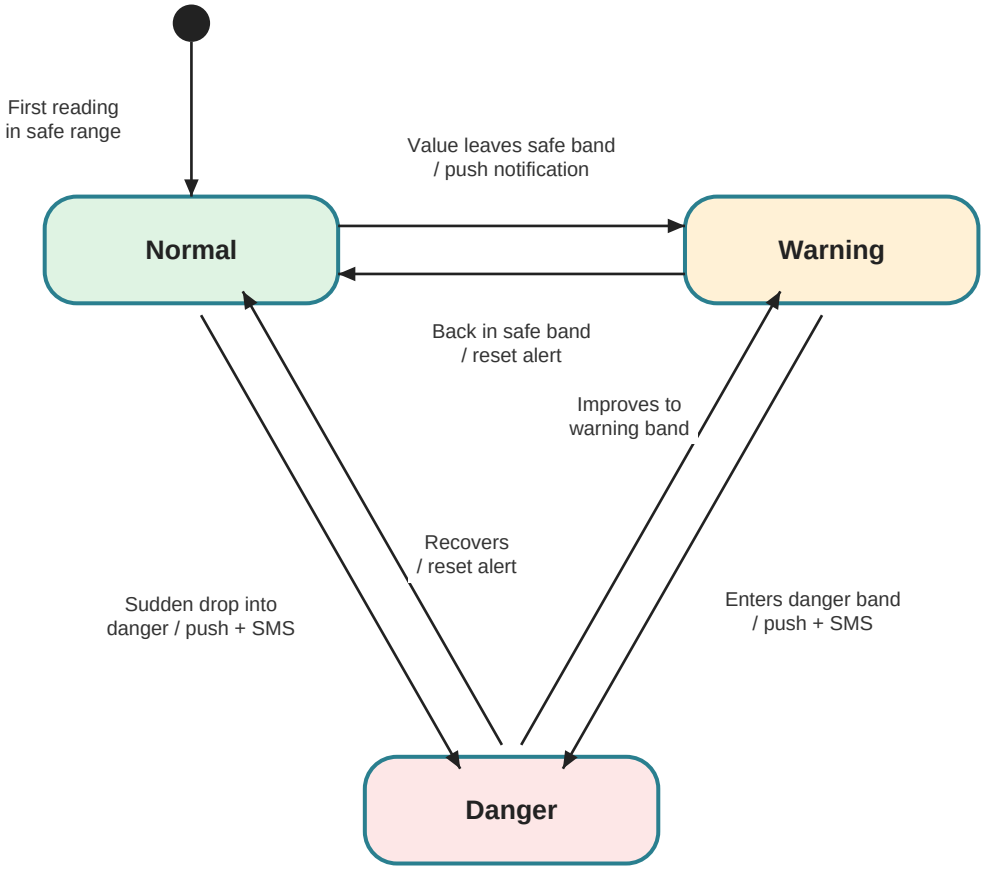
4. State Transition Diagrams

Left: life cycle of a pond's ownership verification. Right: life cycle of one water metric's alert status.

A. Pond Verification



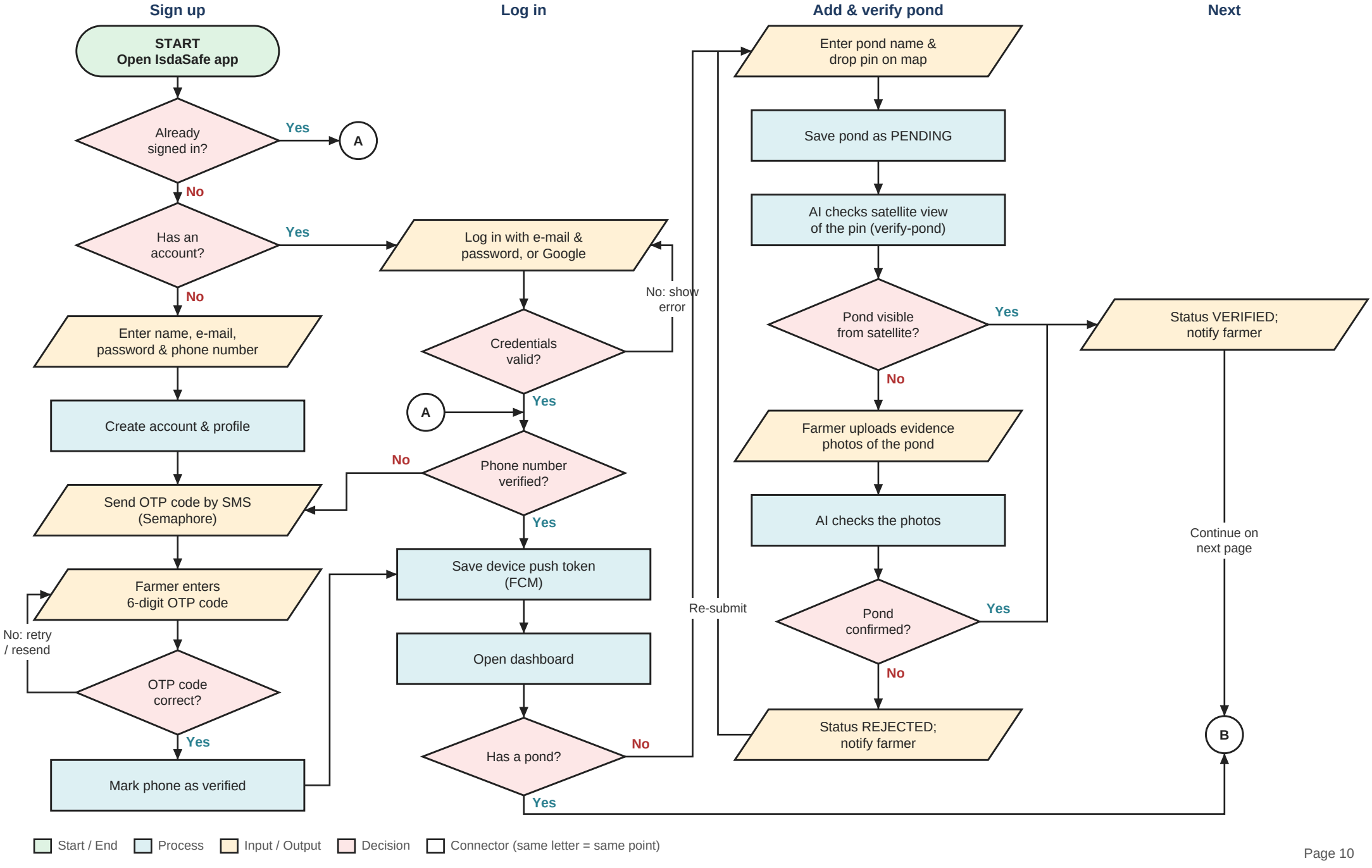
B. Water Metric Alert Status



Event / action notation: "trigger / what the system does".
An alert is only sent when the status rises above the last alerted level, so the farmer is not spammed while a metric stays bad.

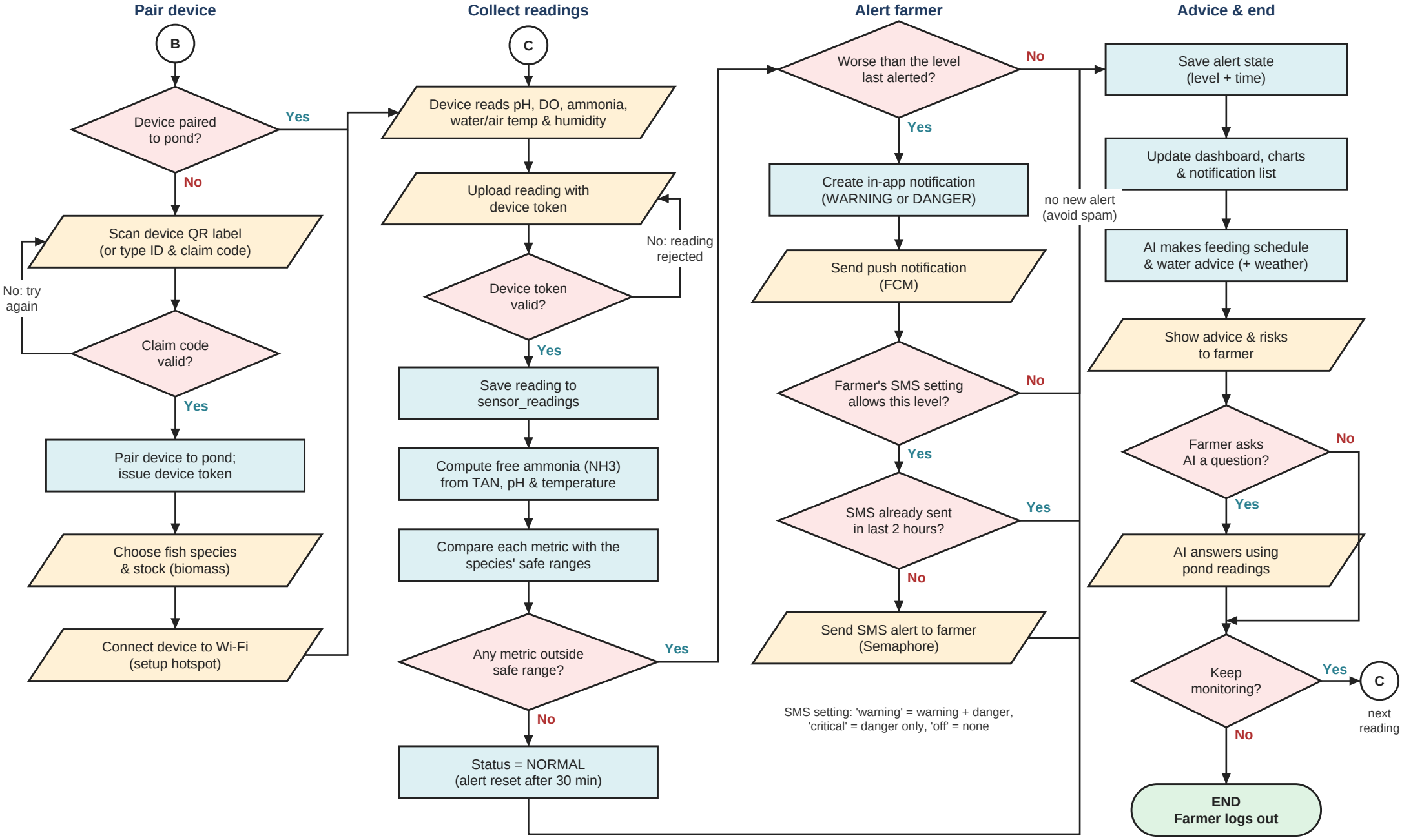
5. Flowchart (1 of 2) - Account, Log In & Pond Verification

The whole IsdaSafe system from opening the app to logging out. Read top to bottom, left to right.



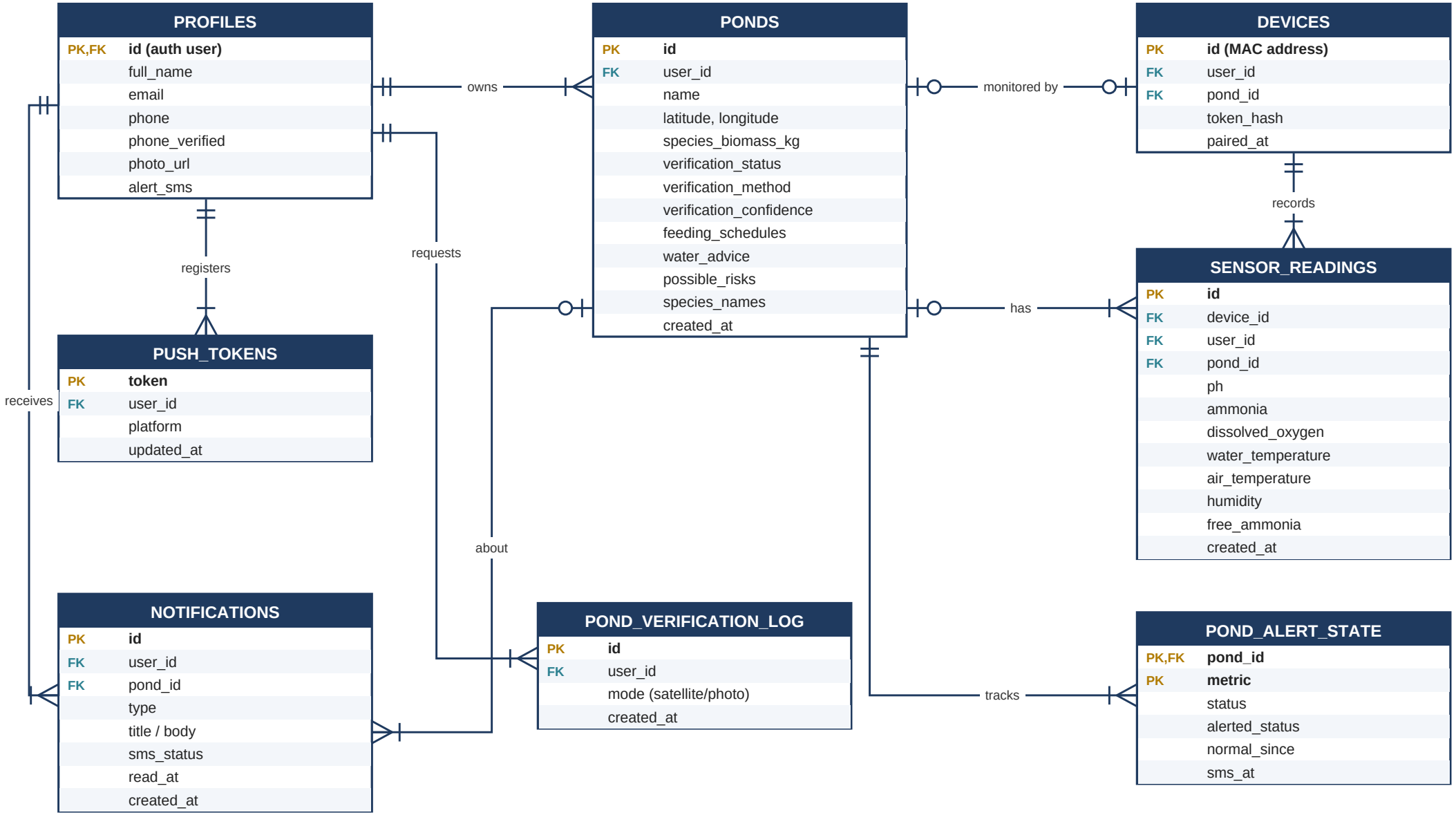
5. Flowchart (2 of 2) - Device Pairing, Monitoring, Alerts & AI Advice

The whole IsdaSafe system from opening the app to logging out. Read top to bottom, left to right.



6. Entity Relationship Diagram

Main IsdaSafe tables in Supabase (PostgreSQL). Crow's-foot notation: | = one, < = many, o = optional. Key columns only - see the Data Dictionary for every column.



7. Data Dictionary

All tables in the public schema of the IsdaSafe Supabase database. PK = primary key, FK = foreign key, timestamptz = timestamp with time zone.

Table 7.1 profiles

Account details of each registered fish farmer (one row per login account).

Column	Data type	Null	Key	Default	Description
id	uuid	No	PK, FK	-	User ID; references auth.users.id (deleted with the account)
full_name	text	Yes		-	Farmer's full name
email	text	Yes		-	Login e-mail address
phone	text	Yes		-	Mobile number used for OTP and SMS alerts
phone_verified	boolean	No		false	True once the phone number passed OTP verification
created_at	timestamptz	No		now()	Date and time the profile was created
photo_url	text	Yes		-	URL of the profile picture (avatars storage bucket)
alert_sms	text	No		'warning'	Which water alerts are texted: 'warning' (warning + danger), 'critical' (danger only) or 'off'

Table 7.2 devices

IoT sensor devices paired to a farmer account and (optionally) a pond.

Column	Data type	Null	Key	Default	Description
id	text	No	PK	-	Device ID = 12-character MAC address (0-9, A-F)
user_id	uuid	No	FK	-	Account that owns the device; references auth.users.id
pond_id	uuid	Yes	FK	-	Pond the device is installed in; references ponds.id (set null if pond deleted)
token_hash	text	No	UNIQUE	-	Hash of the secret token the device uses to upload readings
paired_at	timestamptz	No		now()	When the device was paired to the account
wifi_setup_requested_at	timestamptz	Yes		-	Owner asked the device to open its Wi-Fi setup hotspot; cleared when picked up
wifi_ssid	text	Yes		-	Name of the Wi-Fi network the device is connected to
wifi_ssid_changed_at	timestamptz	Yes		-	When the Wi-Fi network last changed
last_seen_at	timestamptz	Yes		-	Last time the device contacted the server (online status)

Table 7.3 device_credentials

Factory credentials of every manufactured device, used to verify QR claims.

Column	Data type	Null	Key	Default	Description
id	text	No	PK	-	Device MAC address (12 hex characters)
secret_hash	text	No	UNIQUE	-	Hash of the device secret stored in the firmware
claim_code_hash	text	No		-	Hash of the claim code printed on the device QR label
created_at	timestamptz	No		now()	When the credential was registered

7. Data Dictionary (continued)

All tables in the public schema of the IsdaSafe Supabase database. PK = primary key, FK = foreign key, timestamptz = timestamp with time zone.

Table 7.4 ponds

Fish ponds registered by farmers, with verification results and AI-generated advice.

Column	Data type	Null	Key	Default	Description
id	uuid	No	PK	gen_random_uuid()	Unique pond ID
user_id	uuid	No	FK	-	Owner of the pond; references auth.users.id
name	text	No		-	Pond name given by the farmer
latitude	double precision	Yes		-	Latitude of the pond pin on the map
longitude	double precision	Yes		-	Longitude of the pond pin on the map
species_names	text[]	No		'{}'	Fish species stocked in the pond (e.g. Tilapia, Bangus)
created_at	timestamptz	No		now()	Date and time the pond was added
verification_method	text	Yes		-	How the pond was verified: 'satellite' or 'photo'
verification_confidence	real	Yes		-	Confidence score (0-1) returned by the AI verification
photo_path	text	Yes		-	Storage path of the pond photo
photo_hash	text	Yes		-	Hash of the pond photo (detects re-used images)
verification_status	text	No		'verified'	'pending', 'verified', 'rejected', 'error' or 'needs_photos'
verification_message	text	Yes		-	Explanation shown to the farmer about the verification result
verification_started_at	timestamptz	Yes		-	When the current verification run started
evidence_photo_paths	text[]	No		'{}'	Storage paths of extra evidence photos (1-6) when photos are requested
evidence_photo_hashes	text[]	No		'{}'	Hashes of the evidence photos
feeding_schedules	jsonb	No		'{}'	AI-generated feeding schedule per species
water_advice	jsonb	No		'{}'	AI-generated water level / water quality recommendations
possible_risks	jsonb	No		'{}'	AI-generated list of possible water risks
species_biomass_kg	jsonb	No		'{}'	Estimated fish biomass in kg per species (must be a JSON object)

Table 7.5 device_claim_log

Log of device claim attempts (used to rate-limit QR claiming).

Column	Data type	Null	Key	Default	Description
id	bigint	No	PK	identity	Auto-increment log ID
user_id	uuid	No	FK	-	User who tried to claim; references auth.users.id
device_id	text	No		-	MAC address of the device being claimed
created_at	timestamptz	No		now()	When the attempt happened

7. Data Dictionary (continued)

All tables in the public schema of the IsdaSafe Supabase database. PK = primary key, FK = foreign key, timestamptz = timestamp with time zone.

Table 7.6 device_transfer_requests

Requests to move a device that is already owned to another account.

Column	Data type	Null	Key	Default	Description
id	uuid	No	PK	gen_random_uuid()	Unique request ID
device_id	text	No		-	MAC address of the device to transfer
requester_id	uuid	No	FK	-	User asking for the device; references auth.users.id
owner_id	uuid	No	FK	-	Current owner who must approve; references auth.users.id
status	text	No		'pending'	'pending', 'approved', 'declined' or 'cancelled'
created_at	timestamptz	No		now()	When the request was made
resolved_at	timestamptz	Yes		-	When the request was approved, declined or cancelled

Table 7.7 sensor_readings

Water-quality measurements uploaded by the IoT devices.

Column	Data type	Null	Key	Default	Description
id	bigint	No	PK	identity	Auto-increment reading ID
device_id	text	No		-	MAC address of the device that sent the reading (devices.id)
dissolved_oxygen	real	Yes		-	Dissolved oxygen, mg/L
ph	real	Yes		-	pH level (0-14)
ammonia	real	Yes		-	Total ammonia nitrogen (TAN), mg/L, from the SEN0711 sensor
water_temperature	real	Yes		-	Water temperature, degrees C
air_temperature	real	Yes		-	Air temperature, degrees C
humidity	real	Yes		-	Relative humidity, %
created_at	timestamptz	No		now()	When the reading was recorded
user_id	uuid	Yes	FK	-	Owner of the device at reading time; references auth.users.id
pond_id	uuid	Yes	FK	-	Pond measured; references ponds.id (set null if pond deleted)
free_ammonia	real	Yes		-	Un-ionized ammonia NH3, mg/L, computed from TAN, pH and temperature (Emerson 1975)

Table 7.8 push_tokens

Firebase Cloud Messaging tokens of the phones/browsers where the app is installed.

Column	Data type	Null	Key	Default	Description
token	text	No	PK	-	FCM registration token
user_id	uuid	No	FK	-	Account logged in on that phone/browser; references auth.users.id
platform	text	No		'android'	Platform of the token (android, web, ...)
updated_at	timestamptz	No		now()	Last time the token was refreshed

7. Data Dictionary (continued)

All tables in the public schema of the IsdaSafe Supabase database. PK = primary key, FK = foreign key, timestamptz = timestamp with time zone.

Table 7.9 pond_alert_state

Current alert level of each water metric per pond (prevents repeated alerts).

Column	Data type	Null	Key	Default	Description
pond_id	uuid	No	PK, FK	-	Pond; references ponds.id (deleted with the pond)
metric	text	No	PK	-	Water metric name, e.g. ph, dissolved_oxygen, free_ammonia
status	smallint	No		0	Latest status: 0 = normal, 1 = warning, 2 = danger
alerted_status	smallint	No		0	Level the farmer was last notified about (0 = none outstanding)
normal_since	timestamptz	Yes		-	When the metric last returned to normal
sms_at	timestamptz	Yes		-	When the last SMS about this metric was sent
sms_status	smallint	No		0	Alert level of that last SMS

Table 7.10 notifications

In-app notifications shown to the farmer (water alerts, verification, device events).

Column	Data type	Null	Key	Default	Description
id	uuid	No	PK	gen_random_uuid()	Unique notification ID
user_id	uuid	No	FK	-	Recipient; references auth.users.id
pond_id	uuid	Yes	FK	-	Related pond; references ponds.id (set null if pond deleted)
type	text	No		-	Kind of notification, e.g. water_alert_*, pond_needs_photos, device_transfer_request
title	text	No		-	Notification title
body	text	No		-	Notification message
sms_status	text	Yes		-	'sent', 'failed', 'skipped', 'pending', 'not_configured' or 'rate_limited'
read_at	timestamptz	Yes		-	When the farmer opened it (null = unread)
created_at	timestamptz	No		now()	When the notification was created

Table 7.11 phone_otps

Active one-time password for phone-number verification (one per user).

Column	Data type	Null	Key	Default	Description
user_id	uuid	No	PK, FK	-	User verifying a phone; references auth.users.id
phone	text	No		-	Phone number the OTP was sent to
code_hash	text	No		-	Hash of the OTP code
expires_at	timestamptz	No		-	When the OTP stops being valid
attempts	integer	No		0	Number of wrong guesses so far
locked	boolean	No		false	True after too many wrong attempts

7. Data Dictionary (continued)

All tables in the public schema of the IsdaSafe Supabase database. PK = primary key, FK = foreign key, timestamptz = timestamp with time zone.

Table 7.12 otp_send_log

Log of every OTP SMS sent (used to rate-limit OTP requests).

Column	Data type	Null	Key	Default	Description
id	bigint	No	PK	identity	Auto-increment log ID
user_id	uuid	No		-	User who requested the OTP
phone	text	No		-	Phone number the OTP was sent to
created_at	timestamptz	No		now()	When the OTP was sent

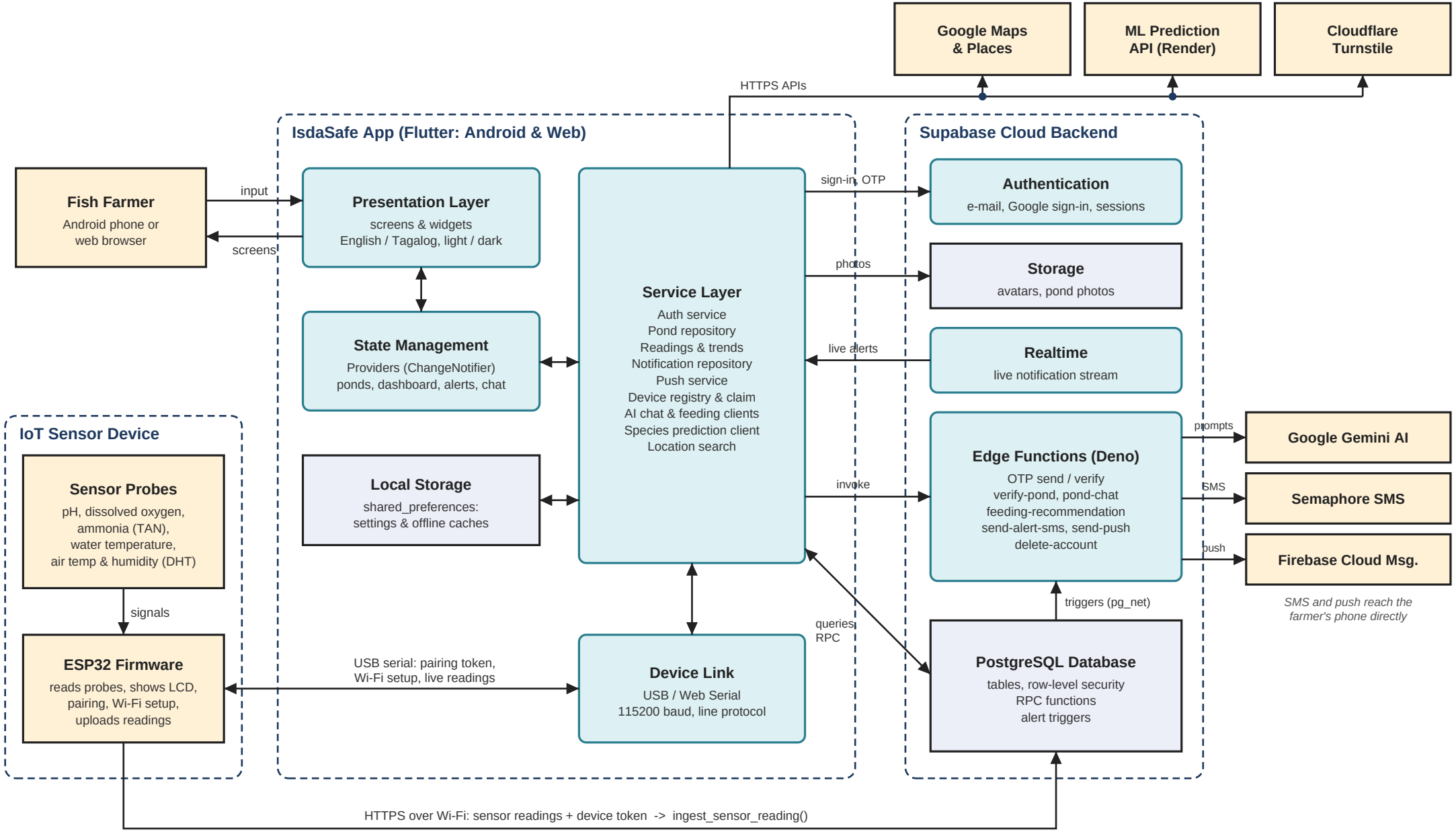
Table 7.13 pond_verification_log

Log of pond verification runs per user (used to rate-limit verification).

Column	Data type	Null	Key	Default	Description
id	bigint	No	PK	identity	Auto-increment log ID
user_id	uuid	No	FK	-	User who ran the verification; references auth.users.id
mode	text	No		-	'satellite' or 'photo'
created_at	timestamptz	No		now()	When the verification ran

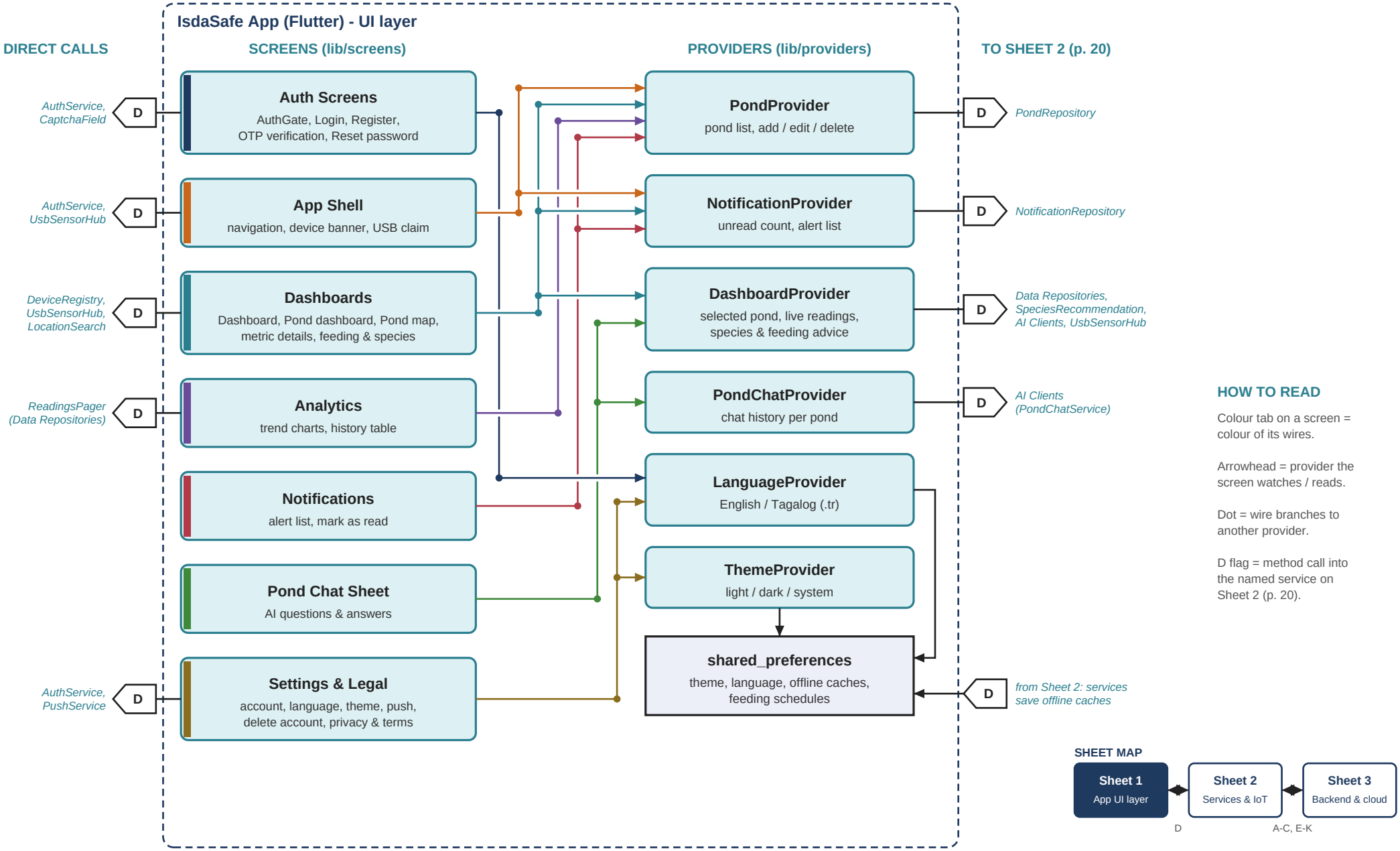
8. Software Block Diagram

The IsdaSafe software as its major blocks and the main data passed between them. Dashed outlines group the blocks that run on the same platform.



9. Software Schematic (Sheet 1 of 3) - App Screens & Providers

Each screen has its own colour: follow it to the arrowheads to see which providers it uses. D flags are calls into the named service on Sheet 2 (page 20).



HOW TO READ

Colour tab on a screen = colour of its wires.

Arrowhead = provider the screen watches / reads.

Dot = wire branches to another provider.

D flag = method call into the named service on Sheet 2 (p. 20).

SHEET MAP

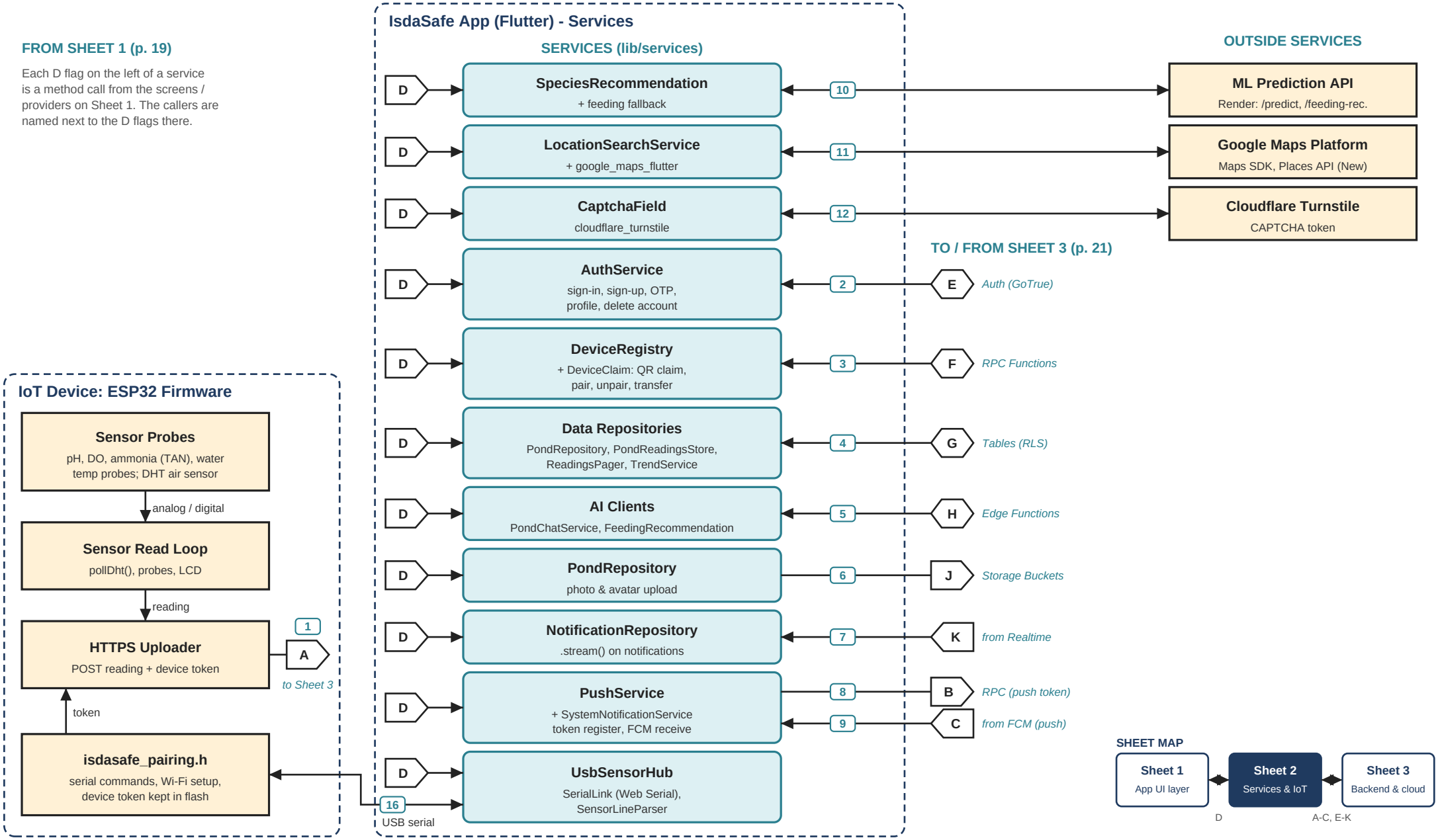
Software module | Data store | Platform | Screen wire (colour per screen) | D Off-page connector (same letter on another sheet)

9. Software Schematic (Sheet 2 of 3) - App Services & IoT Device

App services, the ESP32 firmware and the outside APIs the app calls directly. Connector D comes from Sheet 1 (p. 19); A-C and E-K continue on Sheet 3 (p. 21).

FROM SHEET 1 (p. 19)

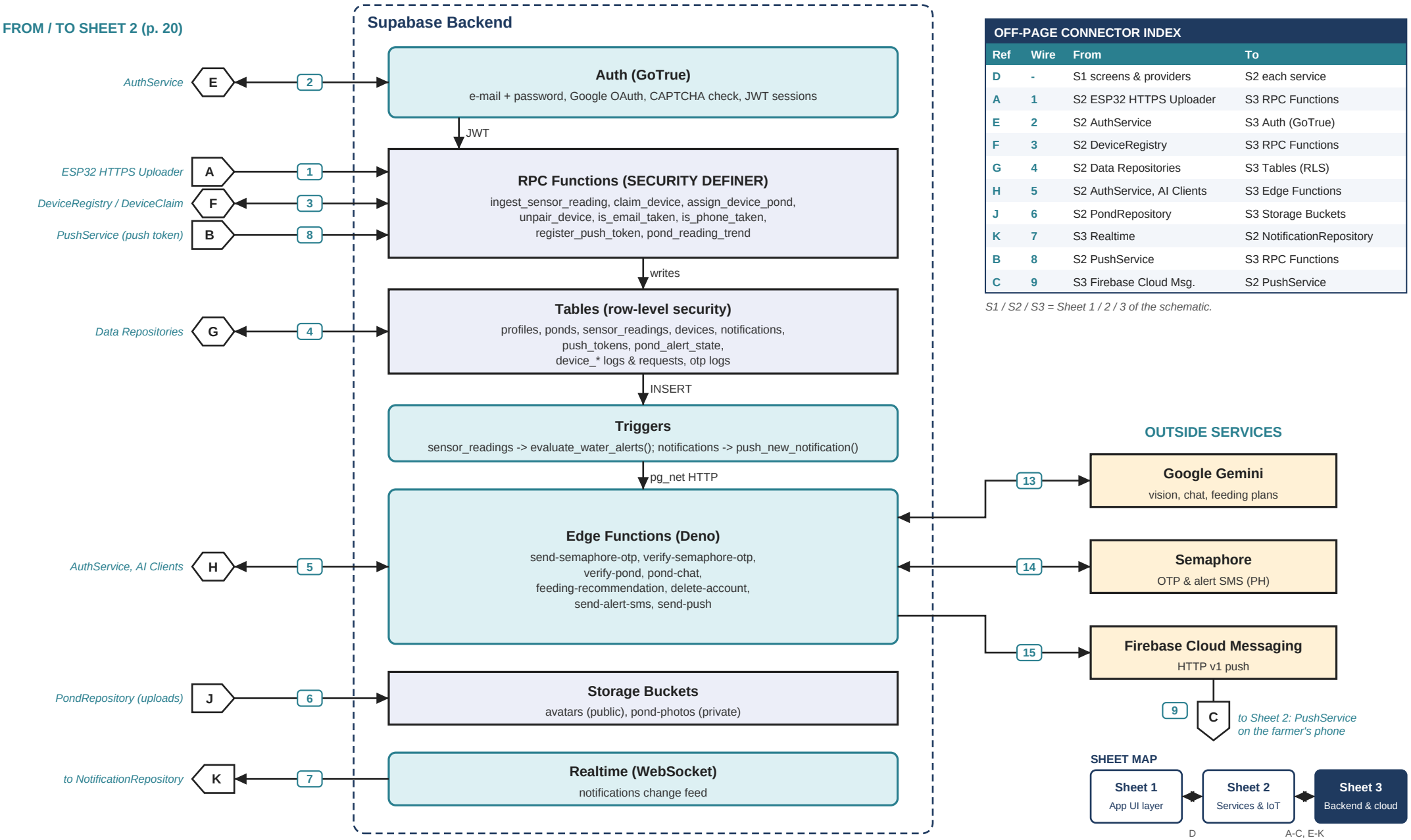
Each D flag on the left of a service is a method call from the screens / providers on Sheet 1. The callers are named next to the D flags there.



Hardware / outside service · Software module · Platform · n Wire no. (Interface List) · D Off-page connector (same letter on another sheet)

9. Software Schematic (Sheet 3 of 3) - Backend & Cloud

Every wire entering from the left comes from Sheet 2 (p. 20) through the off-page connector with the same letter. Numbers match the Interface List on the next page.



OFF-PAGE CONNECTOR INDEX

Ref	Wire	From	To
D	-	S1 screens & providers	S2 each service
A	1	S2 ESP32 HTTPS Uploader	S3 RPC Functions
E	2	S2 AuthService	S3 Auth (GoTrue)
F	3	S2 DeviceRegistry	S3 RPC Functions
G	4	S2 Data Repositories	S3 Tables (RLS)
H	5	S2 AuthService, AI Clients	S3 Edge Functions
J	6	S2 PondRepository	S3 Storage Buckets
K	7	S3 Realtime	S2 NotificationRepository
B	8	S2 PushService	S3 RPC Functions
C	9	S3 Firebase Cloud Msg.	S2 PushService

S1 / S2 / S3 = Sheet 1 / 2 / 3 of the schematic.

Outside service | Software module | Data store | Platform | n Wire no. (Interface List) | D Off-page connector (same letter on another sheet)

9. Software Schematic Diagram (continued) - Interface List

Each numbered wire of the schematic (Sheets 1-3): the two modules it joins, the interface or protocol used, and the data carried.

#	From	To	Interface / protocol	Data carried
1	ESP32 HTTPS Uploader	RPC ingest_sensor_reading	HTTPS POST (PostgREST), anon key + device token	pH, DO, ammonia, water / air temp, humidity
2	AuthService	Supabase Auth	supabase_flutter (HTTPS, JWT)	sign-up / sign-in, Google OAuth, sessions, CAPTCHA token
3	DeviceRegistry, DeviceClaim	RPC functions, devices table	PostgREST RPC (HTTPS)	QR claim code, pond assignment, unpair, transfer requests
4	Data repositories	Tables (RLS)	PostgREST select / insert / update	ponds, sensor_readings, devices, trend buckets
5	AuthService, AI clients	Edge Functions	functions.invoke (HTTPS + JWT)	OTP send / verify, pond chat, feeding advice, delete account
6	PondRepository, AuthService	Storage buckets	Storage API (HTTPS upload)	pond evidence photos, profile avatar
7	Realtime	NotificationRepository	WebSocket (Realtime .stream())	new / updated notifications for the signed-in user
8	AuthService, PushService	RPC functions	PostgREST RPC (HTTPS)	is_email_taken, is_phone_taken, register / unregister push token
9	Firebase Cloud Messaging	PushService (phone)	FCM push (Android), Notifications API (web)	water alert, verification and device notifications
10	Species & feeding clients	ML Prediction API (Render)	HTTPS POST JSON	latest readings -> recommended species, feeding plan
11	LocationSearchService, map	Google Maps Platform	Places API (New) + Maps SDK	search text -> places; map tiles for pond pins
12	CaptchaField	Cloudflare Turnstile	Turnstile widget (WebView / JS)	one-time CAPTCHA token for login / register
13	verify-pond, pond-chat, feeding	Google Gemini	HTTPS (generativelanguage API)	satellite / pond photos, chat prompts, pond context
14	OTP & alert SMS functions	Semaphore	HTTPS (api.semaphore.co)	OTP codes, warning / danger water alerts
15	send-push	Firebase Cloud Messaging	FCM HTTP v1 (OAuth service account)	push payload to every registered token
16	UsbSensorHub (SerialLink)	isdasafe_pairing.h	USB serial / Web Serial, 115200 baud	pairing token, Wi-Fi SSID & password, live reading lines

Wires between screens, providers and services are in-app Dart calls (Provider context.watch / read and async method calls) and are not numbered.